# AI Image Generator

## 6. Models and Prompt Enhancement

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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, Qwen-Image, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and an Admin panel that manages models, LoRAs and downloads and shows live system statistics.

# 6. Models and Prompt Enhancement

## 6.1 Built-in models

Every model is described by a ModelSpec in models.py: where it comes from, its family, its default steps and guidance, the sampler it runs with, the style words that auto-select it and a one-line description.

| Say | Model | Family | Steps | CFG | Sampler | Best for |
|---|---|---|---|---|---|---|
| model zimage | Z-Image Turbo | zimage | 8 | 0 | flow matching (shipped) | default model: photoreal, text rendering and strong prompt following in 8 steps |
| model sdxl | SDXL base 1.0 | sdxl | 30 | 7 | DPM++ 2M Karras | general purpose baseline |
| model juggernaut | Juggernaut XL v9 | sdxl | 30 | 5 | DPM++ 2M Karras | photoreal people, products and scenes; the best all-round upgrade |
| model realvis | RealVisXL V5.0 | sdxl | 30 | 4 | DPM++ SDE Karras | photoreal with strong skin, fabric and lighting detail |
| model dreamshaper | DreamShaper XL v2 Turbo | sdxl | 6 | 2 | DPM++ SDE Karras | fast illustration, fantasy and stylised art |
| model playground | Playground v2.5 | sdxl | 40 | 3 | EDM (shipped) | highly aesthetic, painterly-real composition and colour |
| model animagine | Animagine XL 3.1 | sdxl | 28 | 6 | Euler a | anime and manga characters |
| model qwen | Qwen-Image + Lightning | qwen | 8 | 1 | flow matching, exponential shift | 20B model in 4-bit: best text rendering and layout, posters and signage |
| model flux | FLUX.1 schnell | flux | 4 | 0 | flow matching (shipped) | best prompt understanding: text, counting, complex scenes |

Aliases also work: z-image, zit, jugg, realistic, turbo, aesthetic, anime-xl, qwen-image, schnell. Playground keeps its own EDM scheduler because its weights were trained with it; Z-Image, Qwen-Image and FLUX use flow-matching schedulers.




*Figure 6.1: The same prompt ("close-up portrait of an old fisherman with weathered skin, golden hour light, photorealistic, seed 11") rendered by the six SDXL-family models at 832x1216 with their default settings.*

## 6.2 Automatic selection

When the Model box is on Auto and the prompt names no model, the loaded model is kept while it is Z-Image Turbo or FLUX, because both handle every style themselves. When an SDXL-family model is loaded the style preset decides: photorealistic and cinematic choose Juggernaut, anime chooses Animagine, oil painting and watercolor choose Playground, digital art, comic and cyberpunk choose DreamShaper. The switch happens only if that model is already on disk; otherwise the loaded model is kept and the notes say which model to request. With no style word the loaded model is reused, so consecutive prompts never pay a swap unless they ask for one.

## 6.3 Steps and guidance follow the model

| Prompt says | SDXL (30 / 7) | DreamShaper Turbo (6 / 2) | FLUX (4 / 0) |
|---|---|---|---|
| nothing | 30 steps, cfg 7 | 6 steps, cfg 2 | 4 steps |
| quick | 15, 7 | 3, 2 | 2 |
| highly detailed or 4k | 40, 7 | 8, 2 | 5 |
| strict | 30, 9.1 | 6, 2.6 | 4 (note: no guidance) |

| Prompt says | SDXL (30 / 7) | DreamShaper Turbo (6 / 2) | FLUX (4 / 0) |
|---|---|---|---|
| creative | 30, 4.9 | 6, 1.4 | 4 (note: no guidance) |
| 25 steps cfg 3 | 25, 3 | 25, 3 | 25, 3 |

## 6.4 Prompt weighting and long prompts

- `(glowing sword:1.3)` or `(glowing sword)++` emphasises; `(word:0.7)` or `(word)--` de-emphasises. Automatic1111-style syntax is converted to compel's.
- Prompts longer than 77 tokens are encoded in chunks instead of being cut off. Both positive and negative embeddings are padded to the same length with the empty-prompt embedding.
- Applies to every SDXL-family model, including local checkpoints. FLUX receives the plain text; its T5 encoder reads 256 tokens natively.
- If compel ever fails on a prompt, the pipeline's own encoder is used and a line is printed; generation never fails because of weighting.

## 6.5 LoRAs

Two starter SDXL LoRAs are set up by `python models.py fetch-loras` (they are not kept in git): **Pixel Art XL** (nerijs/pixel-art-xl, 163 MB, trigger word "pixel art") and **Papercut** (TheLastBen/Papercut_SDXL, 325 MB, trigger word "papercut"). Tick them in the LoRAs box under the negative prompt, where they apply at weight 0.8, or write the tag in the prompt with any weight.

- Files in models/loras/ are matched by name, ignoring case and punctuation: `<lora:Pixel_Art_XL:0.9>` finds pixel-art-xl.safetensors.
- LoRAs trained for SDXL apply to every SDXL-family model. Z-Image and FLUX accept LoRAs trained for them; an SDXL LoRA requested on those models is reported as incompatible in the notes and skipped.
- Each LoRA is loaded into the resident pipeline once and then activated or deactivated per request with its weight; swapping models reloads them on demand.
- LoRAs patch the UNet and the text encoders, so they interact correctly with weighting.
- Weights are clamped to the range -2 to 2.

## 6.6 Local checkpoints

Any single-file SDXL checkpoint in models/checkpoints/ is loaded with diffusers' from_single_file. The key is the file name in lower-case kebab form, the description shows its size, and turbo, lightning or hyper in the name selects 6 steps, guidance 2 and the SDE sampler. The folder is scanned at start-up.

## 6.7 Z-Image Turbo, the default model

Z-Image Turbo (Tongyi-MAI, Apache 2.0) is a 6-billion-parameter single-stream flow transformer with a Qwen3 text encoder, distilled to run in 8 steps without classifier-free guidance. It renders photoreal scenes and legible text, follows long prompts closely, and understands English and Chinese. The app loads it at start-up and uses it whenever the prompt names no model and no style word points at a specialist. On a 24 GB card the loader enables per-model CPU offload so the text encoder leaves the GPU after encoding and the 12 GB transformer has room for activations. Like FLUX it has no negative prompt and no guidance scale, and LoRAs are not applied to it; prompt weighting is also not available because its text encoder is not CLIP.

## 6.8 Qwen-Image on 24 GB

Qwen-Image (Alibaba, Apache 2.0) is a 20-billion-parameter MMDiT with a Qwen2.5-VL-7B text encoder and the strongest open-weight text rendering and layout control. Its bf16 weights are 41 GB, so on a 24 GB card the loader quantises the transformer to 4-bit NF4 shard by shard straight onto the GPU on first use and saves the result under models/cache/; the text encoder comes pre-quantised from the diffusers NF4 repository. The

Lightning 8-step LoRA from lightx2v is attached as a permanent adapter with the exponential time-shift scheduler it was distilled for, which replaces the stock 50 steps with classifier-free guidance by 8 steps without it. User LoRAs stack on top of Lightning. The model has no negative prompt in this configuration and the guidance word scaling is ignored.

## 6.9 FLUX.1 schnell on 24 GB

FLUX.1 is a 12-billion-parameter flow-matching transformer. Its bf16 weights alone are 24 GB, so on a card under 40 GB the loader quantises the transformer to 4-bit NF4 and the T5 text encoder to 8-bit with bitsandbytes, then enables per-model CPU offload. Expect a few minutes of load time and about 15 to 25 s per 1024x1024 image. The repository is gated: the loader raises a clear error with the login instructions if the terms have not been accepted. This path is implemented but has not been exercised on the development machine because no Hugging Face token was available.